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A Thesis Report

On

ARTIFICIAL INTELLIGENCE FOR CYBER SECURITY THREATS

Submitted to Governor's State University, USA

In Partial Fulfillment of the requirements for the Degree of

MASTER'S IN INFORMATION TECHNOLOGY

Submitted

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ARTIFICIAL INTELLIGENCE FOR CYBERSECURITY THREATS

Abstract - As the digital landscape becomes more intricate, cyber threats continue to advance in sophistication and scale, demanding proactive and adaptive solutions. This thesis delves into the realm of Artificial Intelligence (AI) and its transformative impact on bolstering cybersecurity defenses against a myriad of threats. By harnessing AI's capabilities, ranging from machine learning algorithms to deep neural networks, this study explores the potential to significantly enhance threat detection, response, and mitigation strategies.

This study also addresses the need for human-machine collaboration, recognizing the integral role of skilled cybersecurity professionals in conjunction with AI algorithms. The findings contribute to a deeper understanding of AI's capabilities and limitations, providing insights that can guide the development and deployment of AI-driven security systems. We look at real examples of AI success and consider ethics and teamwork with human experts. The findings show AI's promise in making cybersecurity stronger against changing cyber threats.

Keywords-Cyberattacks, AI, Cyber threats, Vulnerability, Response, Detection.

1. INTRODUCTION

The term cybersecurity refers to a set of technologies, processes, and practices to protect and defend networks, devices, software and data from attack, damage, or unauthorized access [1]. Cybersecurity is becoming complex because of the exponential growth of interconnected devices, systems, and networks. This is

exacerbated by advances in the digital economy and infrastructure, leading to a significant growth of cyberattacks with serious consequences. In addition, researchers report the continued evolution of nation-state-affiliated and criminal adversaries, as well as the increasing sophistication of cyberattacks, which are finding new and invasive ways to target even the savviest of targets [2]. This evolution is driving an increase in the number, scale, and impact of cyberattacks and necessitating the implementation of intelligence-driven cybersecurity to provide a dynamic defense against evolving cyberattacks and to manage big data. Advisory organizations, such as the National Institute of Standards and Technologies (NIST), are also encouraging the use of more proactive and adaptive approaches by shifting towards real-time assessments, continuous monitoring, and data-driven analysis to identify, protect against, detect, respond to, and catalogue cyberattacks to prevent future security incidents [3].

Artificial Intelligence was developed in the 20th century. This development resulted from trying to create a structure that would not require the help of a human brain. The discovery led to more research being conducted on the matter [8]. More people have tried to create intelligent systems and robots. The developments all attempted to include an object that mimics human behavior and acts without significant impact on humans. The research was also included in mathematics, where several mathematicians tried to develop formulas to help with the aspect. Organizations poured much money to ensure these research studies were successful. The entire history of AI

showcases the growth that the technology has come. AI platforms assist enterprises in the development, management, and deployment of machine learning and deep learning models at scale. Decreasing software development tasks such as data management and deployment make AI technology more accessible and economical [9]. With the increase in cyber risks, artificial intelligence (AI) is increasingly widely employed to monitor and restrict cybercrime.

AI is an intriguing tool that can provide analytics and intelligence to protect against ever evolving cyberattacks by swiftly analyzing millions of events and tracking a wide variety of cyber threats to anticipate and act in advance of the problem. For this reason, AI is increasingly being integrated into the cybersecurity fabric and used in a variety of use cases to automate security tasks or support the work of human security teams. The flourishing field of cybersecurity and the growing enthusiasm of researchers from both AI and cybersecurity have resulted in numerous studies to solve problems related to the identification, protection, detection, response, and recovery from cyberattacks.

Several reviews on cybersecurity and AI applications were published in recent years [4–7]. However, to the best of our knowledge, there is no comprehensive review that covers state-of-the-art research to explain cybersecurity activities covered by AI techniques and the details of how they are applied. Therefore, our objective was to provide a systematic review, a comprehensive view of AI use cases in cybersecurity, and a discussion of the research challenges related to the adaptation and use of AI for cybersecurity to serve as a reference for future researchers and practitioners.

Research Objectives:

The primary objective of this research is to explore the transformative impact of AI on cybersecurity defenses. This study aims to:

- Investigate the role of AI, including machine learning and deep neural networks, in enhancing threat detection capabilities.
- Analyze AI-driven response strategies to cyber threats, with a focus on real-time adaptability.
- Assess the necessity and potential of human-machine collaboration in the field of cybersecurity.
- Examine ethical considerations surrounding the use of AI in cybersecurity and propose ethical guidelines.
- Provide insights and recommendations for the development and deployment of AI-driven security systems to address evolving cyber threats.

Through a comprehensive examination of AI's capabilities, limitations, and real-world successes, this research strives to contribute valuable knowledge to the field of cybersecurity, ultimately making cybersecurity defenses more resilient in the face of ever-changing cyber threats.

2. LITERATURE REVIEW

Background and Context:

This section is dedicated to analyzing the background information concerning the key concepts of this review, including the operational definition of cybersecurity using the NIST cybersecurity framework [3] and the AI taxonomy proposed by AI Watch to clarify the concept of different applications of AI for cybersecurity.

2.1. Cybersecurity

Cybersecurity puts policies, procedures, and technical mechanisms in place to protect, detect, correct, and defend against damage, unauthorized use or modification, or exploitation of information and communication systems and the information they contain. The rapid pace of technological change and innovation, along with the rapidly evolving nature of cyber threats, further complicates the situation. In response to this unprecedented challenge, AI-based cybersecurity tools have emerged to help security teams efficiently mitigate risks and improve security. Given the heterogeneity of AI and cybersecurity, a uniformly accepted and consolidated taxonomy is needed to examine the literature on applying AI for cybersecurity. This structured taxonomy will help researchers and practitioners come to a common understanding of the technical procedures and services that need to be improved using AI for the implementation of effective cybersecurity.

For this purpose, a well-known cybersecurity framework proposed by NIST was used to understand the solution categories needed to protect, detect, react, and defend against cyberattacks [3]. The NIST cybersecurity framework's core describes the practices to improve the cybersecurity of any organization. The framework's core has four elements: Functions, Categories, Subcategories, and Informative references. The first two levels of the NIST framework, which consist of 5 cybersecurity functions and 23 solution categories, were used to classify the identified AI use cases. The functions provide a comprehensive view of the lifecycle for managing cybersecurity over time. The solution categories listed under each function offer a good starting point to identify the AI use cases to improve the cybersecurity. The main purpose of selecting

these two levels is to provide a clear and intuitive categorization to classify the existing AI for cybersecurity literature into the appropriate solution category. The proposed taxonomy introduces a third level consistent with the first two levels by specifying AI-based use cases corresponding to each level of the cybersecurity framework, as shown in Fig. 1.

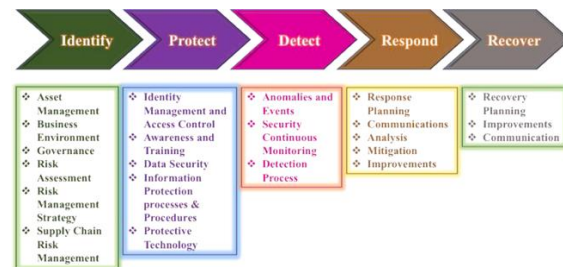


Fig. 1. NIST Cybersecurity Framework

2.2. Artificial intelligence

Several definitions of AI systems can be found that relate to the fields in which they are used and the stages of an AI system's life-cycle, such as research, design, development, deployment, and use. Since the focus of this paper is on AI applications for cybersecurity, a pre-vailing, but simplified, definition of AI is adopted: "systems that exhibit intelligent behavior by analyzing their environment and with some degree of autonomy take actions to achieve specific goals" [9]. In practical terms, AI refers to a number of different technologies and applications that are used in a variety of ways. AI use cases in cybersecurity describe which environmental situations are desirable and undesirable and assign actions to sequences.

For this SLR, the AI taxonomy proposed by Samoil et al. [8], which defines the core and transversal AI domains and subdomains, is used. The core AI domains, i.e., reasoning, planning, learning, communications, and

perception, were found to be useful as they encompass the main scientific areas of AI. Reasoning deals with knowledge representation and different ways of reasoning, while planning also covers searching and optimization. Learning includes machine learning; communication is related to natural language processing; and perception is about computer vision and audio processing [8]. The approaches and technologies that make up these AI domains include, but are not limited to, fuzzy logic, case-based reasoning, genetic algorithm, Bayesian optimization, evolutionary algorithm, planning graph, artificial neural network, deep learning, support vector machine, natural language processing, text mining, sentiment analysis, image processing, sensor networks, object recognition and speech processing.

AI is a large, multidisciplinary research area, with a large body of literature addressing its applications and consequences from a variety of perspectives, e.g., technical, operational, practical, and philosophical. This study focuses on the literature's thread that discusses the implications of the aforementioned methods and AI applications in cybersecurity scenarios. It analyses in detail how AI methods can be used for the identification, protection, detection, response, and recovery in the domain of cybersecurity.

In recent years, the digital landscape has witnessed an unprecedented proliferation of cyber threats. These threats, ranging from sophisticated malware and ransomware attacks to data breaches and identity theft, have not only grown in frequency but have also evolved in complexity. The consequences of successful cyberattacks can be devastating, leading to significant financial losses, reputational damage, and even national security concerns. As

organizations and individuals become increasingly reliant on digital technologies, the need for robust cybersecurity measures has never been more pressing.

The traditional approaches to cybersecurity, while effective to some extent, are often reactive in nature. They rely on known patterns and signatures of threats, leaving organizations vulnerable to emerging and unknown threats. To address this ever-evolving landscape, the integration of Artificial Intelligence (AI) has emerged as a transformative paradigm in cybersecurity.

In today's digital age, the relentless growth of technology and interconnectedness has led to a corresponding increase in the complexity and frequency of cyber threats. The cyberspace, once heralded for its transformative potential, has become a battleground where malicious actors exploit vulnerabilities for financial gain, espionage, and disruption of critical services. As organizations and individuals continue to rely heavily on digital infrastructure, the need for robust cybersecurity measures has never been more pressing.

In response to these evolving threats, the field of cybersecurity has witnessed significant advancements in techniques and tools. Artificial Intelligence (AI), particularly machine learning and deep neural networks, has emerged as a transformative force with the potential to revolutionize cybersecurity. AI's ability to process vast datasets, detect anomalies, and adapt in real-time aligns seamlessly with the dynamic nature of cyber threats.

3. RESEARCH OBJECTIVES

Investigating the Role of AI in Enhancing Threat Detection:

In recent years, the proliferation of cyber threats has necessitated a more proactive and intelligent approach to threat detection. Traditional rule-based systems are often inadequate in identifying emerging threats, as they rely on predefined signatures and patterns. AI, particularly machine learning (ML) and deep neural networks (DNNs), offers a paradigm shift in this regard [10].

Machine Learning for Threat Detection:

Machine learning algorithms excel in recognizing complex patterns and anomalies within vast datasets, a crucial capability in cybersecurity. They can process diverse types of data, including network traffic, logs, and user behavior, to discern subtle deviations from normal patterns. ML models can identify outliers and potential threats that might go unnoticed by rule-based systems. For instance, ML-based intrusion detection systems (IDS) can learn from historical data and adapt to new attack techniques. They can distinguish between normal network behavior and suspicious activities, reducing false positives and enhancing the accuracy of threat detection.

Deep Neural Networks (DNNs) in Cybersecurity:

Deep learning, a subset of ML, focuses on neural networks with multiple layers (DNNs). DNNs have demonstrated remarkable success in various cybersecurity applications, such as:

- *Malware Detection:* DNNs can analyze the binary code of software to identify malicious patterns and classify malware accurately.
- *Phishing Detection:* They can examine email content, URLs, and sender behavior to detect phishing attempts, protecting users from fraudulent messages.

- *Anomaly Detection:* DNNs excel at identifying unusual behaviors within networks, endpoints, and user activities. They can spot zero-day attacks and novel threats [10].

Real-time Threat Detection:

One of the key advantages of AI, especially deep learning, is its ability to provide real-time threat detection. DNNs can process data streams rapidly, allowing organizations to respond swiftly to emerging threats, minimizing potential damage.

Adaptive Threat Detection:

AI systems continuously learn from new data, adapting to evolving threat landscapes. They can detect novel attack techniques and zero-day vulnerabilities by recognizing deviations from learned models. This adaptability is crucial in an environment where cyber threats constantly evolve.

Reducing False Positives:

AI-powered threat detection systems are known for their ability to reduce false positives. By analyzing contextual information and behavior patterns, they can accurately distinguish between benign anomalies and genuine threats, allowing security teams to focus on critical issues.

Analyze AI-driven response strategies to cyberthreats:

The urgency to fortify defenses against the weaponization of Artificial Intelligence (AI) in cyberattacks has never been greater. Adversaries can now harness AI to launch targeted cyberattacks with unprecedented precision and exploit vulnerabilities at speeds and scales unattainable by human hackers.

AI can also be used to craft highly convincing phishing emails, create malware that adapts to security measures, and even automate the

extraction of valuable data from compromised systems [11].

Expanding threat landscape:

The threat landscape surrounding AI is expanding at an alarming rate. Between January to February 2023, Darktrace researchers have observed a 135% increase in Novel social engineering attacks corresponding with the widespread adoption of ChatGPT. The dynamic nature of AI-driven attacks often renders static defense mechanisms ineffective. Traditional cybersecurity measures, such as signature-based antivirus software, firewalls, and rule-based intrusion detection systems, struggle to keep pace, highlighting the need for more adaptive and advanced cybersecurity strategies.

As AI systems gain autonomy and sophistication, the threat landscape is transformed, demanding immediate attention to prevent disruptions.

Key Initiatives and Solutions:

AI-Specific Threat Detection: Organizations are developing AI-powered threat detection systems that can identify vulnerabilities unique to AI algorithms. By analyzing patterns in AI behavior and data usage, these systems can spot irregularities that might indicate an attack.

Real-time Monitoring: In the age of AI, real-time monitoring is crucial. Company's collaborative efforts focus on providing businesses with tools to monitor their AI systems continuously. This allows for immediate responses to security threats as they arise, reducing the potential for damage.

Strong Security Community: Through engaging with industry experts and cybersecurity professionals, they facilitate information exchange, best practices, and

threat intelligence sharing. In addition to collaborative initiatives, Organizations organize technical workshops and thought leadership conferences, fostering collective defense strategies. These events empower security professionals and businesses to stay resilient in the face of evolving AI-related security risks, encouraging a proactive and collaborative response within the community [11].

What Can Organizations Do to Tackle AI Threats?

Stay Informed: Keep up to date with the latest developments in AI security threats and solutions. Attend industry conferences, webinars, and seminars to gain insights from experts.

Regular Security Audits: Conduct regular security audits of your AI infrastructure to identify vulnerabilities and weaknesses. Collaborate with AI security experts to ensure comprehensive protection.

Invest in AI-Specific Security: Recognize that AI requires specialized security measures. Consider solutions like those offered by companies to safeguard your AI systems effectively.

Train Your Team: Invest in AI security training for your cybersecurity professionals. Equip them with the knowledge and skills needed to defend against AI-specific threats.

Necessity and potential of human-machine collaboration in the field of cybersecurity:

Human-machine collaboration in the field of cybersecurity is not only necessary but also holds significant potential for enhancing the security posture of organizations and individuals. The ever-evolving threat landscape, coupled with the increasing complexity of IT systems and networks, requires a multifaceted approach to defense. The necessity and potential of human-

machine collaboration in cybersecurity is as follows [12]:

Necessity:

Scale and Complexity of Threats: The cybersecurity landscape is constantly evolving, with threats growing in scale and complexity. Automated attacks, such as botnets and ransomware, are capable of exploiting vulnerabilities faster than human defenders can respond. Human agents alone are insufficient to combat these multifaceted threats.

Volume of Data: The amount of data generated by modern digital systems is immense. Humans cannot manually process and analyze this data effectively. Machine learning and artificial intelligence can handle large datasets to detect anomalies and potential threats.

Timeliness of Threat Response: Cyber threats operate at a speed that often exceeds human reaction times. Automated tools can detect and respond to threats in real-time or near-real-time, mitigating potential damage.

Data Analysis: Machine learning models excel at pattern recognition and data analysis. They can identify subtle anomalies in network traffic, user behavior, or system logs that might be indicative of a breach. This analytical capability enhances early threat identification.

Threat Intelligence: AI systems can gather and analyze threat intelligence from various sources, including open-source data, vendor feeds, and internal logs. This enables proactive threat detection and helps organizations stay ahead of attackers.

Automation of Routine Tasks: Many cybersecurity tasks, such as patch management, vulnerability scanning, and log analysis, are repetitive and can be automated. This allows human security professionals to focus on more strategic and complex tasks.

Augmentation of Human Skills: AI and machine learning can complement human

skills and mitigate cognitive limitations. They assist in processing vast amounts of data, freeing up human analysts to make strategic decisions based on their expertise [12].

Potential:

Advanced Threat Detection: Machine learning models can analyze a wide range of data sources, from network traffic to endpoint behavior, to identify patterns indicative of a breach. This allows organizations to detect and respond to threats rapidly.

Vulnerability Scanning and Patch Management: Automated tools can continuously scan systems for vulnerabilities, prioritize them based on risk, and apply patches where necessary. This reduces the attack surface and helps prevent exploitation.

Phishing and Malware Detection: AI-driven systems are effective at identifying malicious emails, websites, and files, helping to prevent phishing attacks and malware infections.

Predictive Analytics: AI can predict potential threats and vulnerabilities by analyzing historical data and trends. This enables organizations to take proactive measures to prevent attacks.

Incident Response: AI-driven systems can automate incident response workflows, enabling faster containment and mitigation of threats.

User Authentication and Access Control: Biometric and behavioral authentication, combined with AI-driven access control, enhance security by minimizing the risk of unauthorized access.

Cybersecurity Education and Training: AI can be used to develop and deliver tailored cybersecurity awareness training, making employees more vigilant against threats.

Challenges:

False Positives and Negatives: AI can generate false alarms (false positives) and miss actual threats (false negatives). Human oversight is necessary to address these issues.

Ethical Concerns: The use of AI in cybersecurity raises ethical concerns, such as privacy implications and the potential for misuse.

Complexity and Cost: Implementing AI and machine learning systems can be complex and expensive, particularly for smaller organizations.

Skilled Workforce: A skilled workforce is required to manage and oversee AI-driven security systems. There is a shortage of cybersecurity professionals with expertise in these technologies.

Ethical considerations surrounding the use of AI in cybersecurity and ethical guidelines.

In the last few years scholars, government organizations and “big tech” all around the world developed a whole body of ethical guidelines or principles for driving the development and deployment of AI applications. The approach followed has been that of principalism. Even if some of these initiatives raised some criticisms [14, 15] since they have been mostly seen as marketing initiative, or an attempt by private sector institutions to avoid legislation or the creation of binding legal norms, they contributed to individuate and collect a very conspicuous set of requirements and principles. Cowls and Floridi [16] assessing five documents, found 44 AI principles addressed while [17] analyzing 22 guidelines extracted 22 different ethical requirements. Starting from Cowls and Floridi [16] some work has been done for exploiting the substantial overlap between the different sets of principles. The main result in such a direction has probably been the work of the European Commission’s High Level Expert Group on artificial intelligence which proposed four principles [18, 19] which are briefly summarized in the following.

- *Respect of human autonomy:* AI systems should not unjustifiably subordinate, coerce, deceive, manipulate, condition or herd humans. Instead, they should be designed to augment, complement, and empower human cognitive, social, and cultural skills. The allocation of functions between humans and AI systems should follow human-centric design principles and leave meaningful opportunity for human choice.

- *Prevention of harm:* AI systems and the environments in which they operate must be safe and secure. They must be technically robust, and it should be ensured that they are not open to malicious use. Vulnerable persons should receive greater attention and be included in the development, deployment, and use of AI systems

- *Fairness:* Considered in both a substantive and a procedural dimension. The substantive dimension implies a commitment to ensure equal and just distribution of both benefits and costs and ensure that individuals and groups are free from unfair bias, discrimination, and stigmatization. The procedural dimension of fairness entails the ability to contest and seek effective redress against decisions made by AI systems and by the humans operating them.

- *Explicability:* The capabilities and purpose of AI systems need to be openly communicated, and decisions – to the extent possible – explainable to those directly and indirectly affected

Ethical assessment of cybersecurity application.

It is a general believe that current cybersecurity solutions will soon be unable to effectively monitor all the internet traffic and timely detect attack vectors: IoT, cloud and 5G will generate levels of internet traffic in

terms of data volume, velocity, and variety which are becoming increasingly difficult to analyze [20]. In such a scenario, organizations face an urgent need to improve their defense strategy. AI techniques are the best candidate for such a role. On the other hand, it is well known that many cybersecurity measures may contribute to infringe some rights and that the unconstrained development of AI applications can lead to situations in conflict with fundamental human rights. Thus, the coupling between cybersecurity and AI needs to be scrutinized. In this section we show how this kind of analysis can be carried out using the framework previously introduced.

A. Identity and access management systems

AI systems can be successfully used to reduce errors during the authentication phase or equivalently to reduce the rate of impersonation attacks. In these cases, AI/ML algorithms need a lot of data for working properly and this data is obtained by profiling human activities. As noted in [21] these systems may contribute to improve system resilience by tracking and collecting “sensor data and human-device interaction from your app/website. Every touch event, device motion, or mouse gesture is collected” thus leading to create a mass-surveillance effect.

Recommendations for the development and deployment of AI-driven security systems to address evolving cyber threats.

The future of AI in security is likely to be marked by continued innovation and development, as new technologies and techniques are developed to enhance the capabilities of AI-powered security systems. Some potential predictions for the future of AI in security and its impact include:

Increased automation: AI-powered security systems are likely to become more automated, reducing the need for human intervention in the security process. This could lead to more efficient and effective security measures but may also raise concerns around transparency and accountability.

Greater accuracy and reliability: As AI algorithms become more sophisticated and are trained on larger and more diverse datasets, they are likely to become more accurate and reliable in their decision-making. This could lead to better threat detection and prevention but could also raise concerns around the potential for errors or biases [22].

Integration with other technologies: AI-powered security systems are likely to become increasingly integrated with other technologies, such as the Internet of Things (IoT) and cloud computing. This could enable more comprehensive and proactive security measures but may also increase the risk of cyber-attacks and data breaches.

Improved privacy and data protection: As concerns around privacy and data protection continue to grow, AI-powered security systems may be developed to incorporate more advanced privacy-preserving techniques, such as federated learning or homomorphic encryption. This could enable more effective security measures while minimizing the risk of data breaches or privacy violations.

Greater collaboration between industry and government: As the threat landscape evolves and becomes more complex, there is likely to be greater collaboration between industry and government in the development and implementation of AI-powered security systems. This could enable more effective

threat detection and prevention but may also raise concerns around privacy and civil liberties.

Overall, the impact of AI on the future of security is likely to be significant, with potential benefits and risks for individuals, organizations, and society as a whole. It will be important to carefully consider these potential impacts and to develop ethical and legal frameworks to guide the development and use of these technologies in a way that is consistent with ethical and legal norms [23].

4.THE ROLE OF AI IN SECURITY

4.1. *Explanation of How AI Can Enhance Security Measures*

Automated threat detection: AI-powered systems can automatically detect and classify threats, including known and unknown threats, in real-time. By analyzing vast amounts of data from different sources, including network traffic, endpoint devices, and logs, AI-powered systems can detect anomalies and patterns that are indicative of an attack [24].

Advanced threat prediction: AI-powered systems can use predictive analytics to identify potential threats before they occur. By analyzing historical data and identifying patterns, AI systems can predict potential security threats and take proactive measures to mitigate them.

Rapid response: AI-powered systems can respond to threats in real-time, providing an immediate response that is faster and more effective than traditional security measures. By using machine learning algorithms to automate response actions, organizations can minimize the impact of a security breach and prevent further damage.

Continuous learning: AI-powered systems can continuously learn from new data and adapt to evolving threats. By analyzing new threats and identifying patterns, AI systems can adjust their algorithms and improve their accuracy and effectiveness over time.

Enhanced authentication: AI-powered authentication systems can use biometric data, behavioral analysis, and other advanced techniques to provide more secure and accurate authentication. By using machine learning algorithms to analyze user behavior and identify potential anomalies, AI-powered authentication systems can detect fraudulent activities and prevent unauthorized access.

Reduced false positives: By using AI-powered systems to analyze data and identify patterns, organizations can reduce the number of false positives generated by traditional security measures. This enables security personnel to focus on real threats and respond more effectively to security incidents.

In summary, AI can enhance security measures by automating threat detection, predicting potential threats, providing a rapid response to security incidents, continuously learning, and adapting to new threats, enhancing authentication, and reducing false positives. By leveraging the power of AI, organizations can enhance their security posture and better protect themselves against cyber threats [25].

4.2. *Discussion of How AI Can Automate Tasks and Reduce False Positives*

One of the key benefits of AI-powered security systems is their ability to automate tasks and reduce false positives. Traditional security systems can generate a large number of false positives, which can lead to alert fatigue and make it more difficult for security

personnel to identify real threats. AI-powered systems can address this challenge in several ways:

Machine learning algorithms can learn from historical data and identify patterns that are indicative of a threat. This enables the system to distinguish between legitimate and malicious activities and reduce the number of false positives generated.

AI-powered systems can use automation to reduce the workload on security personnel. For example, the system can automatically quarantine infected devices, block malicious traffic, or update security policies to address new threats.

AI-powered systems can use natural language processing and other advanced techniques to analyze and classify security alerts. This can help to reduce the number of alerts that require human intervention and enable security personnel to focus on the most critical threats.

By automating tasks and reducing false positives, AI-powered security systems can help to improve the efficiency and effectiveness of security operations. This enables security personnel to focus on the most critical threats and respond more quickly and accurately to security incidents.

5. LIMITATIONS OF AI IN CYBER SECURITY

From Charles Darwin's theory about Man's devolution, we can learn that man has always tried to ensure that they have perfected how nature treats them. The ability to change what nature offers to favor their activities and survival has always been the objective of humanity in ensuring that they have a better environment to stay in. Getting to the industrial stage of the human revolution, we

can see that they have contributed to ensuring that they extensively utilize the knowledge of machinery that will help them in their day-to-day activities [26]. The idea of physics knowledge and how to use and advance machinery helped humanity entirely replace the animals allowing them in their activities. With the help of the machinery, there were able to ensure that they have improved their product and efficiency in their work. A man comes to learn that machinery is better than humans. Therefore, the goal was to entirely replan making with a machine to have more excellent production and avoid any inconvenience brought by human actions. And by developing the machinery, they could get to the computer technology we have today.

Computer technology has become one of the most widely used technologies today, resulting in many essential elements in life being supported by technology. Therefore, some standards must be implemented in the technology to ensure that the efficiency and the security of the services offered are of concern [26]

The ultimate goal of securing the data is achieved. With the idea that machines are better than humans in everything they have been programmed to do, it is logical to say that they will be best at ensuring their security [27]. This results in introducing Artificial Intelligence technology that ensures the machine's security is excellent. There is no instance where the information can get unintended [28]. The AI system works to ensure that they have assigned all the protocols they are programmed to follow to guarantee the security of the data involved.

The AI feeds into different protocols of data encryption and uses other methods. It can generate a more complex way to solve or encrypt the data. With these different data

encryption protocols, the system can ensure that it is difficult enough to ensure that nobody can decode the data involved in the transaction. AI has served networking companies and other organizations efficiently as data security is more advanced and guaranteed. However, considering that man created the technology, they have faults even though they were designed to reprogram and develop themselves in case of any responsibility. The fact that man created the program gives him a chance to study it and reverse engineer the process involved, thereby putting the security problem at risk of getting into the wrong hands [29]. As an individual, has indeed created the AI technology involved in data security.

One of the most significant limitations of AI is that it is just a computer code programmed to ensure that they have followed the protocols and developed themselves in case of anything. This instance may sound okay as they can develop themselves in case of anything. However, the system is entirely programmed; therefore, anybody can take control of them, and they can be manipulated and used as a weapon. Few lines of code are required to be edited, and then the long work hours may be turned into a weapon that will be used against itself. Therefore, with the appropriate ability and knowledge, AI technology can be used as a weapon that will be used to destroy what it was made to protect. This factor is one of AI's most significant limitations to cyber security [30]. Developers and computer scientists should consider this as they understand the capability of AI technology.

AI systems can also be trained to detect cyber threats and malicious malware, thus making them more effective in cyber security. The increasing number of cyber security attacks has led to AI adoption in cybersecurity. The entire process is to ensure that there is efficiency and accuracy. However, AI is

limited and cannot replace humans since it is only instructed to perform a specific task. At times, it cannot detect virtually indistinguishable threats and hence gets into trouble since it looks like the actual message. AI may also find it difficult to detect threats due to evolving cyber threats. Viruses and malware improve at any given time, and so should the AI system need an improvement for efficiency. Also, the practice of cybersecurity is many compared to cybercriminals, who tend to acquire more information on hacking. Therefore, cybercriminals can create a better threat that artificial intelligence would not detect easily [31]. Although AI saves time for the security team, it also requires human experts for creativity, thus making work easier for them. The limitation calls for the developers to ensure they have equipped the technology with multiple capabilities to handle any crime resulting from their restraints.

On the other hand, we can identify that AI technology is not entirely being used to protect data and ensure data security. We can also have the AI technology that is developed to have it generate and create computer viruses. The complexity included in the technology makes it difficult for an individual to compete with the machine, leading to the bridging of data. With as many powers generated in the AI-generated codes, which are used to develop a computer virus, it makes it entirely possible for it to be super easy to corrupt a database and manipulate data in it. This is another limitation that is essentially not to AI technology as a watchdog of cybercrime but as a participant in cybercrime. This limitation showcases another massive impact of AI in cybersecurity. The complexity of the technology is a limitation of AI technology as not everyone in society knows technology. There is also the fact that it is not a simple task to understand the different models

involved in the technology. Technology being so challenging to use and implement to the total capacity can give criminals a chance to get the system as we cannot operate the approach to the maximum capability. The technology requires much information about its operation, which many individuals do not have. Therefore, organizations are still at risk as they cannot get the system to operate to its best.

Also, because we have complexity in the system, we understand that the technology will cost a lot [32]. Therefore, the cost of implementing the technology is much more expensive. Therefore, not all organizations in the world will be able to access the technology and ensure the security of the data. Therefore, the cost of the technology is also a limitation to the implementation [33, 34]. Even though the organization's information is essential to any organization, the cost of implementing AI technology is much higher, limiting the number of individuals who will use the technology for the safety of their data and information. The system's cost results in few members and organizations using the technology, making it hard to appreciate the technology's ability.

6. CONCLUSION

In conclusion, the integration of Artificial Intelligence (AI) into cybersecurity signifies a pivotal evolution in our approach to safeguarding digital landscapes. The thorough examination of AI's role in cybersecurity, starting from the foundational understanding of cybersecurity frameworks and AI taxonomies, illuminated the intricate dynamics at play. The NIST cybersecurity framework and AI taxonomy provided a structured lens, revealing the multifaceted dimensions of cybersecurity challenges and

the diverse applications of AI in addressing them.

The exploration of AI's contributions to threat detection uncovered its transformative potential. Through machine learning mechanisms, particularly within intrusion detection systems and deep neural networks, AI exhibits a capacity for real-time threat identification, adaptability to evolving attack methods, and a substantial reduction in false positives. This signifies a paradigm shift from traditional methods, potentially enhancing the efficiency and efficacy of cybersecurity measures.

However, the journey through the literature also brought to light the inherent limitations and ethical considerations associated with AI in cybersecurity. The principles outlined by the European Commission's High-Level Expert Group underscored the importance of human autonomy, harm prevention, fairness, and explicability in AI development. As AI continues to play an integral role in security, these principles become crucial guidelines for ensuring responsible and ethical AI applications.

Moreover, the discussion extended to the intricate collaboration between humans and machines. The immense scale and complexity of contemporary threats necessitate a synergistic approach, wherein AI automates routine tasks, processes vast datasets, and augments human capabilities. Yet, challenges such as false positives, ethical concerns, and the demand for a skilled workforce highlight the importance of striking a balance between AI's potential and the need for human oversight.

Looking ahead, predictions for the future of AI in cybersecurity forecast increased automation, heightened accuracy, integration with emerging technologies like the Internet of Things (IoT) and strengthened privacy measures. While these advancements hold immense promise, concerns linger around transparency, accountability, and the

potential encroachment on privacy. The collaborative efforts between industry and government, envisioned as a cornerstone for tackling evolving threats, raise additional considerations related to privacy and civil liberties.

In essence, the intricate interplay between AI and cybersecurity demands a nuanced and adaptive approach. The future presents opportunities for innovation and efficiency, but it necessitates a judicious balance between technological advancement, ethical considerations, and collaborative frameworks. Continuous refinement, vigilant safeguards against potential misuse, and the cultivation of a skilled, ethical workforce are imperative for harnessing the full potential of AI in the dynamic landscape of cybersecurity.

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